QUANT

Permutation, Combination & Probability

Assignment 04

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P & C Assignments 04

1. Find the chance of throwing more than 15 in one throw with 3 dice.

1. $\frac{5}{108}$ 2. $\frac{6}{108}$ 3. $\frac{7}{108}$ 4. $\frac{4}{108}$

2. A manufacturing process requires a part to be machined on three lathes of the same kind and two lathes of another kind, the laths being alternated. What is the probability that if the lathes are arranged at random in a line, the manufacturing procedure will be adhered to?

1. $\frac{1}{10}$ 2. $\frac{1}{20}$ 3. $\frac{3}{10}$ 4. None of these

3. Twelve places in a car park are arranged in a row. Eight cars are arranged in the park at random. What is the probability that four successive places in a row will be empty?

1. $\frac{1}{55}$ 2. $\frac{4}{165}$ 3. $\frac{3}{495}$ 4. $\frac{8}{495}$

4. A shop accepts a consignment of ten radio sets if two of the sets chosen at random prove to be in working condition. What is the probability that the shop will accept a consignment with four defective radio sets?

1. $\frac{2}{3}$ 2. $\frac{6}{10}$ 3. $\frac{5}{9}$ 4. $\frac{1}{3}$

5. A has 3 tickets of a lottery in which there are 3 prizes and 6 blanks. B has one ticket of a lottery in which there is 1 prize and 2 blanks. What is the ratio of the probabilities of success of A to B?

1. 16 : 7 2. 16 : 21 3. 1 : 3 4. 7 : 16

6. In the tossing of a coin, find the probability of throwing head and tail alternately in 3 successive trails.

1. $\frac{1}{8}$ 2. $\frac{7}{8}$ 3. $\frac{1}{4}$ 4. $\frac{3}{4}$

7. From 20 tickets marked with the first 20 numerals, one ticket is drawn at random. Find the chance that it is a multiple of 3 or 5.

1. $\frac{10}{20}$ 2. $\frac{9}{20}$ 3. $\frac{8}{20}$ 4. $\frac{9}{21}$

8. 2 whole numbers are taken at random and multiplied together. Find the probability of the last digit of the multiplication being 1, 3, 7 or 9.

1. $\frac{3}{25}$ 2. $\frac{2}{25}$ 3. $\frac{4}{25}$ 4. $\frac{16}{25}$

9. *A* and *B* sit down to gamble, with the prize being Rs. 1,100. Which is to be won by the player who first throws 6. If A has the first throw, what are the odds of A winning the prize?

1. 6 : 5 2. 5 : 6 3. 6 : 11 4. 5 : 11

10. A and B are two inaccurate mathematicians whose chance of solving a given question correctly are $\frac{1}{8}$ and $\frac{1}{12}$ respectively; if they obtain the same result and if it is 999 to 1 odds against their making the same mistake, find the chance that the result is correct .

1. $\frac{96}{1000}$ 2. $\frac{1}{96}$ 3. $\frac{25}{137}$ 4. $\frac{1}{1000}$

Answer key & Explanations

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1.	1	We can get more than 15 in the following ways: 6, 6, 6 ; 6, 6, 5; 5, 5, 6; 6, 6, 4 So there are $1+3+3+3=10$ ways in which we can get more than 15 out of a total of $6\times6\times6=216$ ways. So prob. is $\frac{10}{216}$ ways. So prob. is $\frac{10}{216}=\frac{5}{108}$.
2.	1	Let the 2 types of machines be A and B. We are looking at one arrangement – ABABA. Total number of arrangements possible are $5!/2!\times3!=10$. So the probability of sequence being got is $\frac{1}{10}$.
3.	1	Four successive places can be had in a row of 12 parking lots in 9 ways. The total number of ways in which we can choose 4 places from the 12 lots is ${}^{12}C_4$ i.e $12\times11\times10\times\frac{9}{4}!=\frac{990}{2}=495$. So probability of 4 in a row empty is $\frac{9}{495}=\frac{1}{55}$.
4.	4	Probability of a randomly chosen set being in working conditions is $\frac{6}{10}$. Probability of next set also being in working condition is $\frac{5}{9}$. So probability of acceptance of the sample is $\frac{6}{10}\times\frac{5}{9}=\frac{1}{3}$.
5.	1	Probability of A not winning is $\frac{{}^{6}C_3}{{}^{9}C_3}$. So prob. that A will win at least one prize is $\frac{16}{21}$. Probability of B winning is $\frac{1}{3}$. So ratio of probability of winning of A to B is $\frac{16}{21}:\frac{1}{3}=16:7$.
6.	3	There are a total of $2\times2\times2=8$ possibilities. Of this we are interested in two – HTH, THT. So prob. of this happening is $\frac{2}{8}=\frac{1}{4}$.
7.	2	There are 6 tickets which are multiples of 3, 4 which are multiples of 5 and one which is a multiple of both 3 and 5 in all 6+4-1=9tickets which fit the criteria. So Prob. is $\frac{9}{20}$.
8.	3	Let us start with a multiplication of two numbers. For 1, the numbers have to be $3\times7, 7\times3, 9\times9, 1\times1$, for 3, the numbers have to be $1\times3, 3\times1, 9\times7, 7\times9$. For 7 the only options are $1\times7, 7\times1, 3\times9, 9\times3$. For 9 options are $1\times9, 9\times1, 3\times3, 7\times7$ There are a total of $10\times10=100$ possibilities.

		Of these only $4+4+4+4=16$ possibilities satisfy our requirement. So the required probability is $\frac{16}{100}=\frac{4}{25}$.
9.	1	Prob. of A getting the prize in the first throw is $\frac{1}{6}$. Prob. of A getting the prize in the second throw is $\frac{1}{6}\times\frac{5}{6}\times\frac{5}{6}=\frac{1}{6\times\left(\frac{5}{6}\right)^2}$. Thus A's chance of winning the prize is the sum of the infinite series $\frac{1}{6}\times\left(1+\left(\frac{5}{6}\right)^2+\left(\frac{5}{6}\right)^4+\ldots\right)$ So odds of A winning the prize is 6 is to 5.
10.		Two cases possible – both are correct – prob. is $\frac{1}{8}\times\frac{1}{12}=\frac{1}{96}$. Both are wrong and have made the same mistake, prob is $\frac{1}{1000}$. So prob. of result being correct is $\frac{\frac{1}{96}}{\frac{1}{96}+\frac{1}{1000}}=\frac{25}{137}$.